

MODULE 9

BASIC INTEGRATOR PROJECT: CONVEYOR BELT CONTROL

Siemens TIA Portal PLC Course

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Level: Intermediate

Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

Introduction

This module integrates the knowledge acquired in the previous modules of the course through the development of a real project: the automatic control of an industrial conveyor belt. This type of system is widely used in production lines, warehouses, packaging plants, and logistics processes.

The project combines detection sensors, three-phase motors commanded through contactors, part counters, and process timers, all programmed in TIA Portal V20 using Ladder (LAD) language. The goal is for the student to be able to design, address, and program a complete system from start to finish, replicating the conditions of a real industrial project.

Description of the Industrial Process

The conveyor belt moves parts from a loading point to a discharge point. A sensor detects the presence of each part, a counter keeps track of the parts processed, and a timer manages the system's run, stop, and wait times.

System Elements

- Three-phase motor driving the belt (commanded by a contactor)
- Inductive or photoelectric sensor for part detection
- Start pushbutton
- Stop pushbutton
- Emergency stop (emergency mushroom button)
- Counter for processed parts
- Delay timer for part ejection / diversion
- Run indicator light and fault indicator light

Input/Output Factor (I/O Factor)

The Input/Output Factor (I/O Factor) is the addressing table that relates each physical field element (sensors, pushbuttons, actuators) to its corresponding address in the PLC CPU. This table is a key document in every automation project, since it serves as the reference used during programming, wiring, and system maintenance.

Element	Type	Address	Description
Start Pushbutton	Digital input	I0.0	Starts the conveyor belt
Stop Pushbutton	Digital input	I0.1	Stops the conveyor belt
Emergency Stop	Digital input	I0.2	Immediately cuts the motor run
Part Sensor	Digital input	I0.3	Detects each part passing by
Limit Switch Sensor	Digital input	I0.4	Detects the part at the discharge point

Element	Type	Address	Description
Fault Reset	Digital input	I0.5	Resets the system after a fault
Belt Motor (Contactor)	Digital output	Q0.0	Commands the motor start
Run Indicator Light	Digital output	Q0.1	Indicates the system is running
Fault Indicator Light	Digital output	Q0.2	Indicates a fault condition
Diversion Solenoid Valve	Digital output	Q0.3	Diverts the part when the count is complete

Correctly defining the I/O Factor before programming avoids addressing errors, makes the program easier for others to read, and speeds up commissioning and later maintenance tasks.

Operating Sequence

When Start (I0.0) is pressed, the belt motor (Q0.0) and the run indicator light (Q0.1) are activated.

Each time the part sensor (I0.3) detects an object, the counter increments its value.

When the programmed number of parts is reached, a wait timer is activated.

Once the time has elapsed, the diversion solenoid valve (Q0.3) is activated to separate the batch.

The counter resets automatically and the cycle repeats.

When Stop (I0.1) or Emergency (I0.2) is pressed, the motor stops immediately, and the fault indicator light (Q0.2) is activated in the case of an emergency.

Using Counters

In TIA Portal, counters allow discrete events to be recorded, such as parts passing in front of a sensor. The most commonly used blocks are:

- CTU (Count Up): increments its value on each input edge
- CTD (Count Down): decrements its value on each input edge
- CTUD (Count Up/Down): allows counting in both directions

Example: a CTU counter accumulates the number of parts detected by sensor I0.3. When it reaches a preset value (for example, 10 parts), an output is activated that triggers the batch diversion timer.

Using Timers

Timers make it possible to control the wait, run, and diversion times within the process. The most commonly used blocks in TIA Portal are:

- TON (Timer On Delay): delays the activation of an output

- TOF (Timer Off Delay): delays the deactivation of an output
- TP (Pulse Timer): generates a fixed-duration pulse

Practical example: when the part count is complete, a 3-second TON is triggered, which, upon finishing, energizes the diversion solenoid valve (Q0.3) for a fixed pulse generated by a TP block.

Ladder (LAD) Programming Logic

The general program logic is described below, as structured in the networks of TIA Portal:

Network 1 - Motor Start and Stop

Normally-open contact of I0.0 (Start) in series with the normally-closed contact of I0.1 (Stop) and the normally-closed contact of I0.2 (Emergency), latched with a normally-open contact of the coil itself, activating output Q0.0 (Motor) and Q0.1 (Run indicator light).

Network 2 - Part Counting

A positive edge of I0.3 (Part sensor) increments the CTU block. The current counter value is compared with the setpoint value (PV = 10) using a comparison instruction.

Network 3 - Diversion Timing

The counter's comparison output enables a TON block. When the timing finishes, a TP block is activated that energizes Q0.3 (Diversion solenoid valve) for a fixed time and resets the counter (CTU reset).

Network 4 - Fault Management

Activation of I0.2 (Emergency) forces output Q0.0 (Motor) to zero and latches the activation of Q0.2 (Fault indicator light), requiring I0.5 (Reset) to restore the system.

Programming Best Practices

- Always prepare the I/O Factor before starting programming
- Use clear symbolic names for inputs and outputs
- Properly latch critical safety outputs
- Document each network with descriptive comments
- Verify counter and timer times in simulation (PLCSIM) before commissioning
- Provide a manual reset procedure after any fault condition

Integrative Practical Exercise

Design the complete conveyor belt program in TIA Portal V20, taking into account the following I/O Factor and the following sequence:

Exercise 1: Program the start and stop of the belt motor with latching and emergency stop.

Exercise 2: Implement a CTU counter that accumulates 15 parts detected by the part sensor.

Exercise 3: When the count is complete, activate a 5-second TON timer followed by a 2-second TP pulse on the diversion solenoid valve.

Exercise 4: Incorporate the fault logic with a fault indicator light and manual reset.

Conclusion

The development of this integrative project makes it possible to consolidate, in a single real system, the concepts of digital inputs and outputs, addressing through the I/O Factor, counters, timers, and latching and safety logic. Correct planning of the I/O Factor, combined with organized programming in TIA Portal V20, forms the fundamental basis for developing reliable, maintainable, and scalable industrial automation projects.